

# ROBERT SIMON FONG

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## RESEARCH INTERESTS

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- Differential Geometry, Information Geometry: theory and applications in Manifold Optimization
- Symplectic Geometry and Geometry of Representational Learning
- Machine Learning and Dynamical Systems: Reservoir Computing
- Black-box Optimization: Bayesian Optimization, Zeroth-order Optimization

## EDUCATION

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### University of Birmingham

*Sept. 2015 - July 2020*

*Ph.D., Computer Science*

Thesis: Stochastic Optimization on Riemannian Manifolds

Supervisors: Prof. Peter Tiño, Prof. Joshua Knowles

### University of Waterloo

*Sept. 2013 - Aug. 2014*

*Master of Mathematics, Computational Mathematics*

Thesis: Optimization with Function Values Only

Supervisor: Prof. Thomas Coleman

### University of Waterloo

*Sept. 2007 - Apr. 2012*

*Bachelor of Mathematics, Pure Mathematics (Honours)*

## ACADEMIC POSITIONS

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### Post-Doctoral Research Fellow, University of Tokyo

*May. 2025 - present*

- **Research:** Neuromorphic Reservoir Computing
- Funded the International Research Center for Neurointelligence (WPI-IRCN) at The University of Tokyo Institutes for Advanced Study

### Research Fellow, University of Birmingham

*Feb. 2021 - present*

- **Research:** Reservoir Computing: Universality and applications on time series forecasting.
- **Teaching:** Mathematical Foundations of Artificial Intelligence and Machine Learning (MSc) [06 32250], Algorithms for Data Science (MSc) [06 32258]
- Funded by Alan Turing Institute; Prof. Peter Tiño's Alan Turing Institute Fellowship  
*Machine Learning in the Space of State-Space Dynamic Models*

### Ph.D. Candidate, University of Birmingham

*Sept. 2015 - Jul. 2020*

- **Research** Manifold Optimization using Differential Geometry, Information Geometry and Simplicial Geometry.
- **Teaching:** MSc/ICY Software Workshop (Java) (1,2), Mathematical Foundations of Computer Science

### Master Student, University of Waterloo

*Sept. 2013 - Aug. 2014*

- **Research:** Derivative-free optimization: Simulated Annealing and Derivative-Free Zeroth-Order surrogate directional search methods. (with Prof. Thomas Coleman)

- **Research:** Computational Algebra: Solving multivariate polynomials using diagonal subgroup of linear group action. (with Prof. George Labahn)
- **Teaching:** Math 106 Linear Algebra for Arts (Co-organized).

**Undergraduate Research Assistant**, University of Waterloo

*Jan. 2012 - Apr. 2012*

- Integer Programming and applications on Quay Crane Scheduling problem (special case of Traveling Salesman problem)

**Undergraduate Research Assistant**, Penn State University

*Jan. 2009 - Mar. 2009*

- Optimization of Gravitational Wave Detectors in Laser Interferometer Gravitational-Wave Observatory (LIGO).

## INDUSTRY EXPERIENCE

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**Senior Researcher**, Theory Lab, Huawei

*Aug. 2022 - Mar. 2024*

- Quantum-Inspired Optimization – theory of Hamiltonian-based solver and its applications:
  - Boolean Satisfiability Problem, Electronic Design Automation (EDA), Hardware Verification, Graph partitioning, and MIMO Decoding.
- Reservoir Computing-based Time Series Forecasting
- Hardware Placement and Routing modelling in Integrated Circuits

**Researcher**, Noah's Ark Lab, Huawei

*Aug. 2020 - Aug. 2021*

- Theory of Deep Neural Nets: convexity, modelling, and optimization
- Application and implementation of Deep Neural Nets and Bayesian Optimization:
  - 5G+ High Frequency Surface antenna design and Integrated Circuit Partition.

**Research Analyst**, Cayuga Research Associates

*Jan. 2013 - Aug. 2015*

- Optimization and Simulation of Container Port control system.

**Consultant**, Hong Kong International Terminals Limited

*Apr. 2012 - Jan. 2013*

- Integer Programming Modelling and Abstraction of ISO container placement

## BOOKS AND MONOGRAPHS

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- **Robert Simon Fong** and Peter Tiño. *Population-Based Optimization on Riemannian Manifolds*, volume 1046 of *Studies in Computational Intelligence (SCI)*. Springer, 2022. ISBN: 978-3-031-04292-8 (eBook ISBN: 978-3-031-04293-5)

## SELECTED JOURNAL ARTICLES

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- **Robert Simon Fong**, Gouhei Tanaka, and Kazuyuki Aihara. Symplectic reservoir representation of legendre dynamics, 2025
- **Robert Simon Fong**, Boyu Li, and Peter Tiño. Linear simple cycle reservoirs at the edge of stability perform fourier decomposition of the input driving signals. *Chaos*, 35(4):043109, 2025
- **Robert Simon Fong**, Yanming Song, and Alexander Yosifov. Efficient digital quadratic unconstrained binary optimization solvers for sat problems. *New Journal of Physics*, 27(1):013027, Jan 2025
- Boyu Li, **Robert Simon Fong**, and Peter Tiño. Simple Cycle Reservoirs are Universal. *Journal of Machine Learning Research*, 25(158):1–28, 2024

- Robert Simon Fong. *Stochastic optimization on Riemannian manifolds*. PhD thesis, University of Birmingham, 2020

## SELECTED CONFERENCE PAPERS(REFERRED)

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- Ziqiang Li, **Robert Simon Fong**, Kantaro Fujiwara, Kazuyuki Aihara, and Gouhei Tanaka. Structuring Multiple Simple Cycle Reservoirs with Particle Swarm Optimization. In *2025 International Joint Conference on Neural Networks (IJCNN)*, pages 1–9, 2025
- **Robert Simon Fong**, Boyu Li, and Peter Tino. Universality of real minimal complexity reservoir. *Proceedings of the AAAI Conference on Artificial Intelligence*, 39(16):16622–16629, 2025
- Peter Tiño, **Robert Simon Fong**, and Roberto Fabio Leonarduzzi. Predictive Modeling in the Reservoir Kernel Motif Space. In *2024 International Joint Conference on Neural Networks (IJCNN)*, pages 1–8, 2024
- **Robert Simon Fong** and Peter Tiño. Extended stochastic derivative-free optimization on riemannian manifolds. In *Proceedings of the Genetic and Evolutionary Computation Conference Companion*, GECCO '19, pages 257–258, New York, NY, USA, 2019. ACM
- **Simon Fong** and Peter Tiño. Induced dualistic geometry of finitely parametrized probability densities on manifolds, 2018
- David Tsang, Andrew Lundgren, Ruxandra Bondarescu, **Simon Fong**, and Mihai Bondarescu. Optimizing finite mirrors for advanced gravitational wave detectors. In *APS April Meeting Abstracts*, 2009

## AWARDS AND SCHOLARSHIPS

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|------------------------------------------------------------------------------------------|-------------|
| <b>IRCN Retreat Brainstorming Award FY2025</b> , IRCN, The University of Tokyo           | 2025        |
| <b>Honorary Research Fellowship</b> , University of Birmingham                           | 2021 - 2026 |
| <b>President's Scholarship</b> , University of Waterloo                                  | 2008        |
| <b>AP Scholar with Distinction</b> , The College Board                                   | 2007        |
| <b>Second Honour</b> , Third Pan Pearl Delta plus Chinese Elite Schools Physics Olympiad | 2007        |
| <b>Third Honour</b> , Hong Kong Physics Olympiad                                         | 2007        |
| <b>Third Honour</b> , Hong Kong Physics Olympiad                                         | 2006        |

## INVITED TALKS

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| <b>Drexel University</b> , Philadelphia, PA, USA                       | 28. Feb. 2025 |
| <i>Universality of Simple Cycle Reservoirs and Dilation Theory</i>     |               |
| <b>Invited Talk with Prof. Boyu Li</b>                                 |               |
| <b>Southern University of Science and Technology</b> , Shenzhen, China | 23. Aug. 2024 |
| <i>Universality of Simple Cycle Reservoirs</i>                         |               |
| <b>Invited Talk</b>                                                    |               |

## JP NATIONAL CONFERENCE TALKS

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| <b>JST 若手数学者交流会 (第 7 回)</b> , Tokyo, Japan                      | 21. Mar. 2026 |
| <i>Symplectic Reservoir Representation of Legendre Dynamics</i> |               |
| <b>Poster + Talk</b>                                            |               |
| <b>第 3 回バイオ数理連携会議</b> , Tokyo, Japan                            | 04. Dec. 2025 |
| <i>Simple Cycle Reservoirs: Universality and Applications</i>   |               |
| <b>Poster</b>                                                   |               |
| <b>ICNSM2025</b> , Tokyo, Japan                                 | 19. Nov. 2025 |

*Harmonics at the Edge of Stability*

**Poster**

**IRCN Retreat 2025**, Odawara, Japan

*23. Oct. 2025*

*Simple Cycle Reservoirs: Universality in real numbers*

**Poster + Talk**

**IRCN WPI site visit 2025**, Tokyo, Japan

*16. Jul. 2025*

*Harmonics at the Edge of Stability*

**Poster + Talk**

## RESEARCH GRANTS

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**IRCN Retreat Brainstorming Awards FY2025, JPY 800,000**

*2025-2026*

**Principal Investigator**, 3 partners

*Multi-Reservoir Approach to Modular Invariance and Conservation in Biological Networks*

**AI Incubator Startup Grant, JPY 1,000,000**

*2025-2027*

**Principal Investigator**

*Hamiltonian Representation in Reservoir Computing*

## REVIEW SERVICES

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- SIAM Journal on Optimization
- ACM/SIGEVO Conference on Foundations of Genetic Algorithms (FOGA)